Noncategorifiability of the rank-six fusion ring B_{13}

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Abstract

We exclude an explicit simple, self-dual fusion ring of rank six and multiplicity thirteen from categorification over $\mathbb{C}$, without a pivotal hypothesis. The dimension field of a hypothetical sphericalization is forced to be $\mathbb{Q}(\sqrt{5}, \sqrt{149})$. Character orthogonality, a Casimir trace bound, and formal-codegree divisibility reduce the possible global dimensions to a small integer problem. A necessary norm condition for d-numbers leaves only the rational value 3725. The two complementary signed codegrees must then be roots of $t^2 - 3721t + 3725$, contradicting the d-number theorem. The only exhaustive computation is a bounded integer check; a standard-library Python verifier and all positive candidates are supplied.

1 The ring and the result

The notation B_{13} is a local label for the following ring, not a census identifier. It has basis $1, a, b, c, d, e$, identity duality, and commutative multiplication. Set $P = 12(a + b + c + d + e)$. For nonunit basis elements x, y, the entry in the table is $xy - P$; products involving 1 are fixed by the unit rule.

	a	b	c	d	e
a	$1 + a - 2b - c + e$	$-2a + b + c + d - e$	$-a + b - c$	b	$a - b + e$
b		$1 + a - 3b + c + d + e$	$a + b - 2c - d + e$	$a + b - c + e$	$-a + b + c + d$
c			$1 - a - 2b + c + d$	$-b + c + d$	b
d				$1 + c + d + e$	$b + d + e$
e					$1 + a + d + e$

All structure constants with three nonunit indices lie between 9 and 13. The unit coefficient of xy is $\delta_{x,y}$. The tensor is symmetric in all three indices, so Frobenius reciprocity holds. Every nonunit square contains every basis element; hence the ring is simple. Associativity also follows from the complete character table below, or from the exact tensor check in the verifier.

Theorem 1. *There is no fusion category over $\mathbb{C}$ whose Grothendieck ring, as a based ring, is B_{13}.*

Put $s = \sqrt{5}$, $r = \sqrt{149}$, and $u_\pm = (61 \pm 5r)/2$. Two characters, in the ordered basis $(1, a, b, c, d, e)$, are

$$\chi_\pm = (1, u_\pm - 1, u_\pm, u_\pm - 1, u_\pm + 1, u_\pm + 1).$$

For $v \in \{s, -s\}$, put $p = v - 2$, $h = (v - 1)/2$, and $k = (3 - v)/2$. The other four characters are

$$\rho_v = (1, -p, p, h, -1, k), \qquad \widetilde{\rho}_v = (1, h, p, -p, k, -1).$$

Substitution verifies all character equations. The six rows are mutually orthogonal and nonzero, hence independent; this also proves associativity by identifying the complexified product with coordinatewise multiplication on $\mathbb{C}^6$. The Frobenius–Perron character is χ_+. The formal codegrees are

$$D,\ D',\ 25+10s,\ 25+10s,\ 25-10s,\ 25-10s, \qquad D=\frac{18625+1525r}{2},\quad D'=\frac{18625-1525r}{2}. \tag{1.1}$$

Writing $u=u_+$, we have

$$u^2-61u=1, \qquad D=25ru, \qquad DD'=93125. \tag{1.2}$$

For the fusion matrices $N_i=(N_{ij}^k)_{j,k}$, the Casimir trace is

$$T_0:=\operatorname{Tr}\left(\sum_{i=0}^{5}N_i^2\right)=\sum_{i,j,k=0}^{5}(N_{ij}^k)^2=18725. \tag{1.3}$$

All these identities are checked exactly from the full tensor in `verify.py`.

2 Reduction from an arbitrary categorification

Assume, for contradiction, that $\mathcal{C}$ categorifies B_{13}. Let $\mathcal{S}$ be its finite sphericalization. It has two lifts X_i^+, X_i^- of each simple X_i, interchanged by a sign object ε, with

$$\dim(X_i^-)=-\dim(X_i^+), \qquad \dim(\varepsilon)=-1, \qquad \dim(\mathcal{S})=2\dim(\mathcal{C}).$$

No pivotal structure on $\mathcal{C}$ is needed for this construction [1, Corollary 7.6]. Every simple of $\mathcal{S}$ is self-dual. Indeed, its dual lies over the same self-dual simple of $\mathcal{C}$; if it were the opposite lift, sphericality would equate opposite nonzero simple dimensions. It follows that $K_0(\mathcal{S})$ is commutative.

Choose unit-preserving lifts and form

$$A=K_0(\mathcal{S})/([\varepsilon]+1).$$

This is free of rank six, with basis $x_0=1,x_1,\dots,x_5$. Write $x_ix_j=\sum_k t_{ij}^k x_k$. Then

$$t_{ij}^0=\delta_{ij}, \qquad |t_{ij}^k|\le N_{ij}^k, \qquad t_{ij}^k\equiv N_{ij}^k \pmod 2. \tag{2.1}$$

Its multiplication matrices M_i are real symmetric and commute. Their faithful regular representation is simultaneously diagonalizable, so $A_{\mathbb{Q}}$ is a commutative semisimple algebra, with six real characters.

For any character λ define its signed formal codegree by

$$f_\lambda=\sum_{i=0}^{5}\lambda(x_i)^2.$$

Distinct character rows are orthogonal. The standard coefficient-of-unit pairing on A has the chosen basis as an orthonormal basis; its invariance under multiplication gives this orthogonality, equivalently the simultaneous self-adjointness of the M_i. Consequently

$$\sum_\lambda\frac{1}{f_\lambda}=1, \qquad \operatorname{Tr}\left(\sum_i M_i^2\right)=\sum_\lambda f_\lambda\le 18725. \tag{2.2}$$

For completeness, put $F = \mathrm{diag}(f_\lambda)$ and let V be the character matrix. Orthogonality says $VV^{\mathsf{t}} = F$, so $V^{\mathsf{t}}F^{-1}V = I$. Its unit–unit entry is the first identity. Moreover $f_\lambda > 1$: equality would force every nonunit character value to vanish, contradicting the unit coefficient of x_i^2. Each f_λ is an algebraic integer, being an eigenvalue of the integer matrix $\sum_i M_i^2$.

Set

$$d_i = \dim(X_i^+), \quad d_0 = 1, \qquad K = \mathbb{Q}(d_1, \ldots, d_5), \qquad \Delta = \dim(\mathcal{C}) = \sum_{i=0}^{5} d_i^2.$$

The dimension character factors through A. In particular $[K : \mathbb{Q}] \leq 6$. For any original formal codegree f, the corresponding character pulled back to $K_0(\mathcal{S})$ has formal codegree $2f$. The central-induction theorem [3, Theorem 2.13] gives a central simple of dimension

$$\frac{\dim(\mathcal{S})}{2f} = \frac{\Delta}{f}.$$

Forgetting it to $\mathcal{S}$ shows that

$$f \in K, \qquad \Delta/f \in \mathcal{O}_K. \tag{2.3}$$

Thus $E := \mathbb{Q}(s, r) \subseteq K$. Since $[E : \mathbb{Q}] = 4$ and $[K : \mathbb{Q}] \leq 6$, we obtain

$$K = E = \mathbb{Q}(\sqrt{5}, \sqrt{149}). \tag{2.4}$$

The four embeddings of E therefore give four distinct characters of A, with codegrees $\sigma(\Delta)$. In particular Δ is totally positive.

We also use the d-number theorem [2, Theorem 1.2 and Corollary 1.4]: Δ is a d-number, and so is every f_λ. For the latter assertion, pull λ back through $K_0(\mathcal{S}) \to A$; its codegree is $2f_\lambda$. The Galois ratios of f_λ are the same as those of $2f_\lambda$, so the assertion follows. This step is essential: no positivity assumption on the signed coefficients is being used.

3 Orthogonality and the global-dimension trace

We use the integral basis

$$\mathcal{O}_E = \mathbb{Z}[\phi, \eta], \qquad \phi = (1 + s)/2, \quad \eta = (1 + r)/2. \tag{3.1}$$

Here is a direct maximality check. The product basis has discriminant $5^2\,149^2$, so only 5 and 149 could divide its index in $\mathcal{O}_E$. At 5, the polynomial $t^2 - t - 37$ of η has simple roots $2, 4$. At 149, the polynomial $t^2 - t - 1$ of ϕ has simple roots $41, 109$. Thus at either possible index prime the other quadratic factor splits, and the localized product order is a product of the maximal integer rings in the ramified quadratic extension. It is maximal at both primes. This proves (3.1).

Write, for $1 \leq i \leq 5$,

$$d_i = \frac{\alpha_i + \beta_i s + \gamma_i r + \delta_i sr}{4}, \qquad \alpha_i, \beta_i, \gamma_i, \delta_i \in \mathbb{Z},$$

and put $M = \sum_i \delta_i^2$. Apply orthogonality to the dimension character and each of its three nontrivial Galois conjugates. Taking rational coefficients in the resulting three equations gives

$$16 + \sum_i \alpha_i^2 = 5 \sum_i \beta_i^2 = 149 \sum_i \gamma_i^2 = 745 \sum_i \delta_i^2 = 745M. \tag{3.2}$$

In particular $M > 0$, and

$$\mathrm{Tr}_{E/\mathbb{Q}}(\Delta) = 745M. \tag{3.3}$$

The other two characters of A have codegrees greater than one, so (2.2) gives

$$745M + 2 < 18725, \qquad 1 \le M \le 25.$$

By (1.2) and (2.3),

$$\theta := \frac{\Delta}{25r} \in \mathcal{O}_E. \tag{3.4}$$

Using (3.1), write

$$\theta = \frac{x + ys + mr + wsr}{4}, \qquad x, y, m, w \in \mathbb{Z}. \tag{3.5}$$

Its integrality is equivalent to

$$y \equiv w \pmod 2, \qquad m \equiv w \pmod 2, \qquad x \equiv y + m - w \pmod 4. \tag{3.6}$$

Since $\mathrm{Tr}_{E/\mathbb{Q}}(\Delta) = 3725m$, equation (3.3) gives $M = 5m$. Hence

$$m \in \{1, 2, 3, 4, 5\}, \qquad \Delta = \frac{25}{4}(149m + 149ws + xr + ysr). \tag{3.7}$$

No reduction modulo 2 of the fusion ring and no enumeration of signed associative products is required.

4 A finite norm obstruction

Lemma 2 (Norm–trace test)**.** *Let $F/\mathbb{Q}$ be a Galois number field of degree n, and let $z \in \mathcal{O}_F$ be a nonzero d-number. Then*

$$\frac{\mathrm{Tr}_{F/\mathbb{Q}}(z)^n}{\mathrm{N}_{F/\mathbb{Q}}(z)} \in \mathbb{Z}.$$

This remains true when z generates a proper subfield of F.

Proof. Every $\sigma(z)/z$ is an algebraic unit by [2, Lemma 2.7]. Therefore $\mathrm{Tr}_{F/\mathbb{Q}}(z)/z = \sum_\sigma \sigma(z)/z$ belongs to $\mathcal{O}_F$. Taking its norm proves the assertion. □

For the coordinates in (3.7), total positivity of Δ is equivalent to

$$5w^2 < m^2, \qquad L > 0, \qquad L^2 > 5H^2, \tag{4.1}$$

where

$$L = 149(m^2 + 5w^2) - x^2 - 5y^2, \qquad H = 298mw - 2xy. \tag{4.2}$$

Indeed, for either embedding of $\mathbb{Q}(s)$, positivity at both signs of r requires

$$149(m \pm ws) > r\,|x \pm ys|.$$

Squaring after checking the positive left sides gives (4.1). Adding the two absolute-value bounds also gives

$$|w| \le 2, \qquad |x| < mr \le 5r < 62, \qquad |y| < m\sqrt{149/5} \le \sqrt{745} < 28.$$

Thus the following search bounds are proved:

$$-2 \le w \le 2, \qquad -61 \le x \le 61, \qquad -27 \le y \le 27. \tag{4.3}$$

A relative norm calculation gives

$$n_\theta := \mathrm{N}_{E/\mathbb{Q}}(\theta) = \frac{L^2 - 5H^2}{256} \in \mathbb{Z}_{>0}. \tag{4.4}$$

Since

$$\mathrm{N}_{E/\mathbb{Q}}(\Delta) = 25^4\, 149^2 n_\theta, \qquad \mathrm{Tr}_{E/\mathbb{Q}}(\Delta) = 25 \cdot 149m,$$

Lemma 2 implies

$$n_\theta \mid 149^2 m^4. \tag{4.5}$$

Lemma 3 (Exact bounded arithmetic)**.** *The integer quadruples* (m, w, x, y) *satisfying* (3.6), (3.7), (4.1), *and* (4.5) *consist of the single quadruple* $(4, 0, 0, 0)$. *Consequently* $\Delta = 3725$.

Proof. Use the exhaustive box (4.3). The numbers of integral, totally positive candidates, and the survivors of (4.5), are

m	1	2	3	4	5	total
integral and totally positive	0	66	132	267	664	1129
norm–trace survivors	0	0	0	1	0	1

The following complete integer computation proves these counts and produces the unique surviving quadruple. No numerical approximation is used.

```
from itertools import product
counts = [0] * 6
survivors = []
for m in range(1, 6):
    for w, x, y in product(range(-2, 3), range(-61, 62),
                           range(-27, 28)):
        if 5*w*w >= m*m:
            continue
        if (y-w) % 2 or (m-w) % 2 or (x-y-m+w) % 4:
            continue
        L = 149*(m*m + 5*w*w) - x*x - 5*y*y
        H = 298*m*w - 2*x*y
        if L <= 0 or L*L <= 5*H*H:
            continue
        counts[m] += 1
        numerator = L*L - 5*H*H
        if numerator % 256:
            raise ValueError("Nonintegral norm")
        n = numerator // 256
        if (149**2 * m**4) % n == 0:
            survivors.append((m, w, x, y, n))
print(counts[1:])
print(survivors)
```

The output is

```
[0, 66, 132, 267, 664]
[(4, 0, 0, 0, 22201)]
```

For this survivor, $\theta = r$ and $\Delta = 25r^2 = 3725$. The full verifier independently checks (4.4) by a four-by-four integer determinant for each of the 1129 candidates. □

5 The two remaining codegrees give a contradiction

We isolate the short general observation used in the final step.

Lemma 4 (Two-codegree obstruction)**.** *Let $f, g > 0$ be algebraic integers whose unordered pair is Galois invariant, and suppose $1/f + 1/g = a/b$ in lowest terms, with a, b positive integers. Then*

$$f + g = ka, \qquad fg = kb$$

for a positive integer k. If the polynomial $t^2 - kat + kb$ is irreducible over $\mathbb{Q}$ and f is a d-number, then $b \mid k$.

Proof. The sum $S = f + g$ and product $P = fg$ are positive rational integers, and $S/P = a/b$, so $(S, P) = (ka, kb)$ with $k \in \mathbb{Z}_{>0}$. In the irreducible case, f, g are quadratic conjugates. The d-number condition implies $P \mid S^2$, by Lemma 2 in that quadratic field. Thus $kb \mid k^2a^2$, and coprimality gives $b \mid k$. □

Proof of Theorem 1. By Lemma 3, $\Delta = 3725$. The four dimension characters of A therefore all have codegree 3725. Denote the other two by f, g. The characteristic polynomial of the integer Casimir matrix is

$$(t - 3725)^4(t^2 - St + P), \qquad S, P \in \mathbb{Z},$$

so $\{f, g\}$ is Galois invariant. By (2.2),

$$\frac{1}{f} + \frac{1}{g} = 1 - \frac{4}{3725} = \frac{3721}{3725}.$$

Since $\gcd(3721, 3725) = 1$, Lemma 4 gives

$$S = 3721k, \qquad P = 3725k, \qquad k \in \mathbb{Z}_{>0}.$$

The trace bound in (2.2) gives

$$14900 + 3721k \leq 18725,$$

forcing $k = 1$. Hence f, g are the roots of

$$t^2 - 3721t + 3725. \tag{5.1}$$

Its discriminant 13830941 lies strictly between 3718^2 and 3719^2, so the polynomial is irreducible. But f must be a d-number, whereas

$$3721^2 \equiv 16 \pmod{3725}.$$

This contradicts Lemma 4, equivalently the quadratic d-number condition. Therefore no such $\mathcal{C}$ exists. □

6 Certificate scope and reproduction

The proof imports only sphericalization, the central-induction theorem, and the d-number theorem. Its remaining categorical-algebra reductions are proved above. The finite computation is not a search for associators, signed fusion rings, or pentagon solutions. It is the bounded integer check in Lemma 3.

The accompanying package contains `tensor.json`, `verify.py`, `positive_candidates.csv`, and verification reports. The verifier uses only Python's standard library and checks the full fusion tensor, the six characters and their codegrees, the Casimir trace, the integral basis residue calculations, all bounded arithmetic candidates, an independent norm determinant for each candidate, and the final quadratic contradiction. Run

```
python verify.py --output verification.json \
    --candidates positive_candidates.csv
python -O verify.py --output verification_optimized.json
```

Both commands return `PASS`. Checks use explicit exceptions and are not disabled by `-O`. This verifies the arithmetic certificate; it is not a formalization of the categorical inputs in a proof assistant.

References

[1] P. Etingof, D. Nikshych, and V. Ostrik, *An analogue of Radford's S^4 formula for finite tensor categories*, arXiv:math/0404504.

[2] V. Ostrik, *On formal codegrees of fusion categories*, Math. Res. Lett. **16** (2009), 895–901, arXiv:0810.3242.

[3] V. Ostrik, *Pivotal fusion categories of rank 3*, arXiv:1309.4822.